Review via Linear Regression

02-680: Essentials of Mathematics and Statistics

November 11, 2026

1 Linear Regression

The ***linear regression*** problem is as follows (we covered the least squares solution briefly in Topic 12 (Calculus), written there as $Ax = b$ with x as the unknown; here the unknown is β and x is the data):

x — Independent variable[s] (what we control)

y — Dependent variable[s] (the measured values)

β_s – slope

β_0 – intercept

Some examples of regression in biology:

Independent	Dependent
Expression of genes expressed in immune response	asthma severity
Expression of regulator genes	Expression of target genes
Clinical variables	Insulin level
...	...

Let's assume we have some $x, y \in \mathbb{R}^n$, for n samples where each (x_i, y_i) are corresponding. Then for each sample $y_i = \beta_s x_i + \beta_0$, or for all samples at once

$$y = \beta_s x + \beta_0 \mathbf{1},$$

where $\mathbf{1} \in \mathbb{R}^n$ is the vector of all ones.

To simplify things let's define: $X = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix}$ and $\beta = \begin{bmatrix} \beta_s \\ \beta_0 \end{bmatrix}$, then the equation above

becomes simply:

$$y = X\beta.$$

Recall (possibly) from earlier that we want to find β that minimizes the $\|y - X\beta\|_2^2$, or the square of the L_2-norm.

$$\begin{aligned}
\|y - X\beta\|_2^2 &= (y - X\beta)^T(y - X\beta) \\
&= (y^T - \beta^T X^T)(y - X\beta) \\
&= y^T y + \beta^T X^T X\beta - 2\beta^T X^T y
\end{aligned}$$

To find the critical point of the curve (i.e. the minimum) we need

$$\begin{aligned}
\frac{d}{d\beta} y^T y + \beta^T X^T X\beta - 2\beta^T X^T y &= 0 \\
\frac{d}{d\beta}\beta^T X^T X\beta - \frac{d}{d\beta} 2\beta^T X^T y &= 0 \\
2X^T X\beta - 2X^T y &= 0 \\
2X^T X\beta &= 2X^T y \\
\beta &= (X^T X)^{-1} X^T y
\end{aligned}$$

When does this solution exist?

The last step needs $X^T X$ to be invertible, and eigenelements (Topic 11) tell us when it is.

- $X^T X$ is symmetric, so its eigenvalues are real.
- A matrix is singular exactly when its determinant is 0, and the determinant is the product of the eigenvalues. So $X^T X$ is invertible if and only if ***none*** of its eigenvalues is 0.
- A zero eigenvalue happens when the columns of X are linearly dependent (Topic 10). For example, if every x_i is the same value then the first column of X is a multiple of the column of 1s. With several predictors, the same thing happens if one is a combination of the others (e.g., one measurement is the sum of two others). Then $X^T X\beta = X^T y$ has many solutions and $(X^T X)^{-1}$ does not exist.
- For any eigenvector v of $X^T X$ with eigenvalue λ, we have $\lambda\|v\|_2^2 = v^T X^T Xv = \|Xv\|_2^2 \geq 0$, so every eigenvalue is non-negative. This is why the critical point we found is a minimum of the squared error and not a maximum.

What if I want to know the probability of some y_i given x_i? (so $p(y_i \mid x_i)$)

Assuming it's some natural process, we can assume

$$p(y_i \mid x_i) \sim \mathcal{N}(\beta_0 + \beta_s x_i, \sigma^2).$$

or equivalently

$$y_i = \beta_0 + \beta_s x_i + \epsilon_i, \epsilon_i \sim \mathcal{N}(0, \sigma^2).$$

So given the data, we can actually use MLE to find β! Let $X_i = \begin{bmatrix} x_i & 1 \end{bmatrix}$ be the ith row of X, so the mean of y_i is $X_i\beta = \beta_0 + \beta_s x_i$.

$$\begin{aligned}
\ln p(\mathcal{D} \mid \beta, \sigma) &= \ln \left[\prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(y_i - X_i\beta)^2} \right] \\
&= \sum_{i=1}^{n} \ln \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(y_i - X_i\beta)^2} \right] \\
&= \frac{-n}{2} \ln \sigma^2 - \frac{n}{2} \ln(2\pi) + \sum_{i=1}^{n} \ln e^{-\frac{1}{2\sigma^2}(y_i - X_i\beta)^2} \\
&= \frac{-n}{2} \ln \sigma^2 - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - X_i\beta)^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial\beta} \frac{-n}{2} \ln \sigma^2 - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - X_i\beta)^2 &= \frac{\partial}{\partial\beta} - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - X_i\beta)^2 \\
&= \frac{\partial}{\partial\beta} - \frac{1}{2\sigma^2} \|y - X\beta\|_2^2 \\
&= -\frac{1}{2\sigma^2} \frac{\partial}{\partial\beta} (y^T y - 2y^T X\beta + \beta^T X^T X\beta) \\
&= \ldots
\end{aligned}$$

$\hat{\beta}_{MLE} = (X^T X)^{-1} X^T y$, that's just Least Squares!

But we don't know σ, that's actually, what we want to know because it models how uncertain we are with any given output.

$$\frac{\partial}{\partial \sigma^2} \frac{-n}{2} \ln \sigma^2 - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - X_i\beta)^2 = \frac{-n}{2\sigma^2} + \frac{1}{2}(\sigma^2)^{-2} \sum_{i=1}^{n} (y_i - X_i\beta)^2$$

$\hat{\sigma}^2_{MLE} = \frac{\sum_{i=1}^{n} (y_i - X_i \hat{\beta}_{MLE})^2}{n}$

With these we can determine for some new $\tilde{y}$ and $\tilde{x}$ what $p(\tilde{y} \mid \tilde{x})$ is.